Introduction

Preparedness for incursions of exotic diseases is of key importance to government, industry, producers and the Australian community. Australia's exotic disease preparedness is based on emergency management principles. This includes the development of surveillance, monitoring and early warning systems, the compilation and maintenance of Australia's veterinary emergency plans (AUSVETPLAN) and the conduct of training and awareness programs.

A good understanding of the likely behaviour of exotic disease under Australian conditions is a necessary component of effective preparedness and response planning. In the absence of contemporary Australian experience with such diseases, simulation modelling can be helpful. Simulation provides, in particular, a tool that can be used pre-emptively to investigate the likely spread of disease under different outbreak scenarios and the effectiveness and cost-efficiency of eradication strategies.

The department has been involved with disease modelling for more than 10 years. The objectives of this work have been to identify geographic regions, sub-populations and production systems that might be at greater risk from key exotic diseases such as foot and mouth disease, to evaluate the effectiveness and efficiency of various surveillance and control strategies, to underpin economic impact studies and to provide realistic scenarios for preparedness or training exercises. In agreement with most modelling teams worldwide, the department is cautious about the use of models during an outbreak. In this situation their role should be restricted to the provision of information about the likely effectiveness and efficiency of mitigation strategies under existing outbreak conditions and about resource allocation strategies.

Specific information on some of the modelling projects currently underway can be found at: